

Exact Class-II smart-path cancellation and noncentral shell theorem

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1. Purpose and scope

What closes here. At every finite cutoff this note derives the exact independent-to-self-coupled smart path, integrates the independent carrier, and cancels all apparent trace-class terms by Gaussian integration by parts. It also proves an arbitrary-source, past-measurable frozen-shell determinant/entropy theorem and its summable dyadic Schatten gain.

What remains open. The surviving commutator must still be bounded uniformly under the *tilted interacting laws*. Base-Gaussian moments, L^2 convergence of the A7 composite, and the A8 decoupled determinant do not give that bound. Consequently this note does not prove the self-coupled Nelson estimate or the full three-component Gibbs measure.

The model is the hash-pinned A1 production covariance on $\mathbb{T}_L^3$, $L = 16$, in the six-real convention. The density floor is fixed at $\varepsilon_\rho = 10^{-12}$, Q_{II} is positive, products are exact or sufficiently de-aliased, and every regulator is a common real-even scalar multiplier from the A7 class. Floor removal, asymmetric regulators, and infinite volume are outside the theorem.

1 Finite-cutoff setup

Let E_Λ be the finite white-noise space and write

$$K_\Lambda = R_\Lambda C^{1/2}, \quad G_\Lambda = DK_\Lambda, \quad X = K_\Lambda x, \quad (2.1)$$

where x, y are independent standard Gaussians on E_Λ . The A7/A8 coefficient is

$$B(z) = \sum_{A=1}^3 \{ap_A p_A^T + b(p_A v_A^T + v_A p_A^T) + cv_A v_A^T\} \geq 0, \quad (2.2)$$

with $p_A = 2S_A z$, $v_A = 2(S_A - q_A \mathbf{I})z$, and $q_A = z^T S_A z / (|z|^2 + \varepsilon_\rho)$. Define

$$T(x) = G_\Lambda^* M_{B(K_\Lambda x)} G_\Lambda \geq 0. \quad (2.3)$$

For $p > 0$ and

$$z_s = sx + \sqrt{1 - s^2} y,$$

put

$$H_s(x, y) = U(K_\Lambda x) + \frac{1}{2} \{z_s^T T(x) z_s - \text{Tr } T(x)\}, \quad \Phi_\Lambda(s) = \log \mathbb{E} e^{-pH_s}. \quad (2.4)$$

At $s = 0$ this is exactly the A8 independent-carrier endpoint. At $s = 1$ it is the finite-cutoff A7 self-coupled action.

All derivatives below are ordinary finite-dimensional derivatives. Thus no limiting interchange is hidden in the identities. Write $T_i = DT(x)[e_i]$. Gaussian covariance differentiation gives, for $0 < s < 1$,

$$\Phi'_\Lambda(s) = \mathbb{E}_{\nu_s} \left[p^2 \langle H_x, H_z \rangle - p \operatorname{Tr} H_{xz} \right], \quad (2.5)$$

where ν_s is the tilted probability measure and

$$H_z = Tz, \quad (H_x)_i = U_i + \frac{1}{2}(z^T T_i z - \operatorname{Tr} T_i), \quad \operatorname{Tr} H_{xz} = \sum_i (T_i z)_i. \quad (2.6)$$

Consequently

$$\Phi'_\Lambda(s) = \mathbb{E}_{\nu_s} \left[p^2 \langle DU, Tz \rangle + \frac{p^2}{2} \sum_i (z^T T_i z - \operatorname{Tr} T_i)(Tz)_i - p \sum_i (T_i z)_i \right]. \quad (2.7)$$

Only DB occurs in this first derivative. Higher coefficient derivatives appear only when the centered terms are estimated or integrated by parts.

2 Conditional determinant and exact cancellation

Set

$$t = 1 - s^2, \quad R_t = (I + ptT)^{-1}, \quad S_t = TR_t. \quad (3.1)$$

Integrating y in (2.4) exactly yields

$$\Phi_{\Lambda,t} = \log \mathbb{E}_x e^{W_t(x)}, \quad (3.2)$$

$$W_t = -pU - \frac{1}{2} \log \det {}_2(I + ptT) + \frac{p(1-t)}{2} \{ \operatorname{Tr} T - x^T S_t x \}. \quad (3.3)$$

Here $t = 1$ is the A8 endpoint and $t = 0$ is the A7 endpoint. Direct differentiation gives

$$\partial_t W_t = \frac{p}{2} (x^T S_t x - \operatorname{Tr} S_t) + \frac{p^2(1-t)}{2} \|S_t x\|^2. \quad (3.4)$$

This expression seems dangerous: neither T nor S_t is uniformly trace class. That appearance is false.

Let a_t denote the coefficient-derivative part of $\nabla_x W_t$:

$$(a_t)_i = \frac{p^2 t^2}{2} \operatorname{Tr}(TR_t T_i) + \frac{p(1-t)}{2} \{ \operatorname{Tr} T_i - x^T R_t T_i R_t x \}. \quad (3.5)$$

Since $DS_t[H] = R_t H R_t$,

$$\nabla_x W_t = -pDU - p(1-t)S_t x + a_t. \quad (3.6)$$

Apply Gaussian integration by parts to the vector field $S_t(x)x$:

$$\mathbb{E}_{\nu_t} [x^T S_t x - \operatorname{Tr} S_t] = \mathbb{E}_{\nu_t} [\langle \operatorname{div} S_t, x \rangle + \langle S_t x, \nabla_x W_t \rangle]. \quad (3.7)$$

Substitution of (3.6) into (3.7) cancels the last term in (3.4) exactly and proves the central identity

$$\partial_t \Phi_{\Lambda,t} = \frac{p}{2} \mathbb{E}_{\nu_t} [-p \langle DU, S_t x \rangle + \langle a_t, S_t x \rangle + \langle \operatorname{div} S_t, x \rangle]. \quad (3.8)$$

No estimate was used in this cancellation.

3 Why no Schatten-one term survives

The trace difference in (3.5) has the exact decomposition

$$\mathrm{Tr} T_i - x^T R_t T_i R_t x = \mathrm{Tr}[(\mathrm{I} - R_t^2)T_i] - \{x^T R_t T_i R_t x - \mathrm{Tr}(R_t T_i R_t)\}. \quad (4.1)$$

Moreover

$$\mathrm{I} - R_t^2 = pt S_t (\mathrm{I} + R_t). \quad (4.2)$$

The first term in (4.1) is therefore an $\mathfrak{S}_2$ – $\mathfrak{S}_2$ pairing, while the second is an explicitly centered Hilbert–Schmidt quadratic form. Thus one must not estimate $\mathrm{Tr} T_i$ separately.

There is a second exact cancellation. In coordinates,

$$T_{jk}(x) = \langle G_\Lambda e_j, B(K_\Lambda x) G_\Lambda e_k \rangle_{L^2}. \quad (4.3)$$

The contraction in $\sum_i \partial_i T_{ik}$ contains

$$\sum_i (G_\Lambda e_i)(r) (K_\Lambda e_i)(r)^T = D_r C_\Lambda(r, r). \quad (4.4)$$

Common real-even regulation makes (4.4) zero, hence

$$\mathrm{div} T = 0. \quad (4.5)$$

Using $R_t = \mathrm{I} - pt S_t$ on the left index of $DS_t[e_i]$ gives

$$\mathrm{div} S_t = -pt \mathrm{contr}_i(S_t T_i R_t), \quad (4.6)$$

and therefore

$$\|\mathrm{div} S_t\| \leq pt \|S_t\|_{\mathfrak{S}_2} \left(\sum_i \|T_i\|_{\mathfrak{S}_2}^2 \right)^{1/2}. \quad (4.7)$$

An odd or component-split regulator need not satisfy (4.4). The independent verifier includes such a negative control and confirms that omitting $\mathrm{div} S_t$ then changes the derivative.

At fixed positive floor, $B \in C^\infty$ and the elementary bounds

$$\|B(z)\|_F \leq \beta_B |z|^2, \quad \|DB(z)\| \leq \beta_1 |z| \quad (4.8)$$

hold with finite production constants. The pinned values include

$$\beta_B = 0.256499999999936. \quad (4.9)$$

A8’s spatial trace-ideal estimate applied to $DB(X)[K_\Lambda e_i]$ gives

$$\sum_i \|T_i\|_{\mathfrak{S}_2}^2 \leq C_A \beta_1^2 \kappa_0 \int_{\mathbb{T}_L^3} (1 + |X|)^2 dr, \quad (4.10)$$

where

$$\kappa_0 = \sup_{\Lambda, r} \mathrm{Tr} C_\Lambda(r, r) < \infty \quad (4.11)$$

because the production covariance has order -4 in dimension three. Analogous bounds for differentiated centered quadratics use $D^2 B$ and κ_0^2 . Thus every factor in (3.8) has cutoff-uniform moments under the *base* Gaussian law. Section 7 explains why this is not yet the needed tilted-law estimate.

4 Arbitrary-source frozen-shell theorem

The positive result needed for dyadic freezing is stronger than the A8 zero-source determinant. Condition on all previously exposed shells. Let $b(r) \geq 0$ and a derivative background $a(r)$ be arbitrary measurable functions of that past. For the new standard Gaussian shell ξ_j , put

$$A = M_b^{1/2} G_j, \quad T = A^* A, \quad q = M_b^{1/2} a, \quad (5.1)$$

$$Q = \frac{1}{2} \|q + A\xi_j\|_2^2 - \frac{1}{2} \text{Tr } T. \quad (5.2)$$

Completing the finite-dimensional Gaussian square proves, for every $p > 0$,

$$\begin{aligned} \log \mathbb{E}_j e^{-pQ} &= \frac{1}{2} \{p \text{Tr } T - \log \det(\mathbf{I} + pT)\} - \frac{p}{2} \|q\|^2 \\ &\quad + \frac{p^2}{2} \langle A^* q, (\mathbf{I} + pT)^{-1} A^* q \rangle. \end{aligned} \quad (5.3)$$

The source contribution is nonpositive because

$$pA(\mathbf{I} + pA^*A)^{-1}A^* \leq \mathbf{I}. \quad (5.4)$$

For every eigenvalue $\lambda \geq 0$, $p\lambda - \log(1 + p\lambda) \leq p^2\lambda^2/2$. Hence

$$\boxed{\log \mathbb{E}_j e^{-pQ} \leq \frac{p^2}{4} \|T\|_{\mathfrak{S}_2}^2.} \quad (5.5)$$

The Gibbs variational formula applied with $p = 1/\eta$ gives the equivalent entropy estimate

$$\mathbb{E}_{\nu_j} Q \geq -\eta H(\nu_j \mid \gamma_j) - \frac{1}{4\eta} \|T\|_{\mathfrak{S}_2}^2. \quad (5.6)$$

Equations (5.3)–(5.6) retain an arbitrary low-shell derivative source. This is essential: discarding it would lose the positive energy needed to absorb cross-shell terms.

5 Dyadic gain and harmless independent shifts

Let P_j select a dyadic annulus and $G_j = DP_j R_\Lambda C^{1/2}$. The symbol lower bound

$$A(k) \geq c_{\text{sym}} \langle k \rangle^4 \mathbf{I}, \quad c_{\text{sym}} = 0.108364530587115 \quad (6.1)$$

implies that each differentiated covariance root is bounded by a constant times $\langle k \rangle^{-1}$. Both legs of $T_{j,b}$ lie in the same annulus. Uniformly in the transfer momentum,

$$\sum_{k, k-q \text{ in annulus } j} \langle k \rangle^{-2} \langle k-q \rangle^{-2} \leq C_{\text{sh}}(L) 2^{-j}. \quad (6.2)$$

Indeed the annulus has $O_L(2^{3j})$ modes and each product is $O_L(2^{-4j})$; the finitely many low shells are absorbed into the constant. Therefore

$$\|T_{j,b}\|_{\mathfrak{S}_2}^2 \leq C_{\text{sh}}(L) M_R^4 c_{\text{sym}}^{-2} 2^{-j} \|b\|_{L^2}^2. \quad (6.3)$$

For $b = B(z)$, (4.8) turns (6.3) into a summable multiple of $2^{-j} \|z\|_{L^4}^4$, which the positive sextic can absorb.

There is also an exact sign result. If $h \in H^2$ is deterministic, or is random but independent of X_Λ , same-point common evenness makes $X_\Lambda(r)$ independent of $D_i X_\Lambda(r)$. Conditional on h , the $D_i X$ quadratic term cancels the exact A7 counterterm and the cross term has zero mean. Thus

$$\mathbb{E}_X V_\Lambda^{\text{ren}}(X_\Lambda + h) = \frac{1}{2} \sum_i \int \mathbb{E}_X [(D_i h)^T B(X_\Lambda + h) D_i h] dr \geq 0. \quad (6.4)$$

The obstruction is therefore not an ordinary Cameron–Martin translation; it is genuine same-noise adapted dependence.

6 Exact remaining gate

For one designated dyadic route, let $\phi_j = P_{\leq j}\phi$ and $\Delta_j B = B(\phi_j) - B(\phi_{j-1})$. After applying Section 5 to the frozen part, the unaccounted increment is exactly

$$\mathcal{C}_j = \frac{1}{2} \int (D\phi_j)^T \Delta_j B D\phi_j - \frac{1}{2} \int \text{Tr}[\Gamma_{\leq j} \Delta_j B]. \quad (7.1)$$

A sufficient closure estimate for this route is: for every $\eta > 0$, uniformly in J and every tilted law ν ,

$$\mathbb{E}_\nu \sum_{j \leq J} \mathcal{C}_j \geq -\eta H(\nu \mid \gamma_J) - \eta \mathbb{E}_\nu \|\phi_J\|_6^6 - C_\eta. \quad (7.2)$$

The corresponding smart-path target is a cutoff-uniform bound on the negative variation of (3.8). Either estimate would join the A8 endpoint to the A7 endpoint and prove the desired Nelson bound.

Under the unshifted Gaussian law, power counting gives summable shell increments and the leading cubic commutator has root-mean-square order 2^{-j} . This is not (7.2): relative-entropy tilts may concentrate on rare configurations, and an $L^2(\gamma)$ estimate does not control every such law. The non-exact Class-II K one-form also forbids replacing $B(h)Dh$ by the derivative of a scalar cubic primitive. Positive derivative energy and the noncentral determinant cancellation must be retained.

The umbrella gate remains **A7-CLASSII-SELF-COUPPLING-INTERPOLATION**; its sharp residual after this note is **A7-CLASSII-TILTED-COMMUTATOR-FORM-BOUND**.

7 Devil’s-advocate review

1. **“The raw $\text{Tr} T$ term diverges, so the interpolation fails.” DISMISSED.** Equations (3.8), (4.1), and (4.2) remove it exactly; only Hilbert–Schmidt pairings remain.
2. **“ $\text{div} T = 0$ for every regulator.” UPHELD as false.** It uses the common-even same-point identity (4.4). The executable odd control produces a nonzero correction.
3. **“The frozen-shell theorem only treats zero background.” DISMISSED.** Equation (5.3) includes arbitrary $q = M_b^{1/2} a$, and (5.4) makes its net contribution nonpositive.
4. **“Base-law moments prove the tilted estimate.” UPHELD as false.** This is precisely the distinction between (4.10) and (7.2). The latter remains open.
5. **“Direct Malliavin integration by parts proves the adapted drift bound.” UPHELD as invalid.** For an adapted shift H , the identity contains $\text{Tr}(Db D_x H)$, while the Boue–Dupuis cost controls H but not its Malliavin derivative.
6. **“A new mass or vacuum counterterm has been ruled out at all orders.” UPHELD as an overclaim.** No such term survives the first exact interpolation reduction, but the all-scale tilted commutator audit is unfinished and could still expose one.
7. **“Deterministic lower-bound counterexamples disprove this route.” DISMISSED.** Equation (6.4) shows that deterministic and independent translations are nonnegative after exact covariance centering.
8. **“The dyadic 2^{-j} factor is a fitted slope.” DISMISSED.** It is the analytic mode count (6.2). Executable finite matrices test signs and factors, not the all-shell proof.
9. **“This note constructs the physical Gibbs measure.” UPHELD as false.** The measure remains behind (7.2) or the equivalent tilted smart-path variation bound.

8 Reproduction and references

The fail-closed command is

```
python codes/foundations/a9_classii_smart_path_cancellation_verify.py
```

It runs a primary four-coordinate Fourier model and a non-importing two-coordinate model. Both compare analytic derivatives with direct Gauss–Hermite partition differences, test the noncentral determinant and source sign, check quadrature-error contraction, and include odd-regulator negative controls. The verifier pins all authority hashes and scope text before child execution.

The stochastic-control representation follows M. Boue and P. Dupuis, “A variational representation for certain functionals of Brownian motion,” *Annals of Probability* 26 (1998), <https://doi.org/10.1214/aop/1022855876>. The scale-flow variational strategy is related to N. Barashkov and M. Gubinelli, “A variational method for Φ_3^4 ,” <https://arxiv.org/abs/1805.10814>. These references provide method; the model-specific identities and residual boundary are proved here.

Result footer

Claim: A9-CLASSII-SMART-PATH-CANCELLATION
Tier: T5 CLOSED@EXACT-INTERPOLATION-AND-NONCENTRAL-FROZEN-SHELL
Closed: exact smart path, trace cancellation, noncentral shell theorem
Open: A7-CLASSII-TILTED-COMMUTATOR-FORM-BOUND

Final boundary. Exact interpolation algebra, noncentral frozen-shell control, summable frozen costs, and independent-shift positivity are closed. Uniform tilted-law commutator control, the self-coupled Nelson estimate, the interacting measure, floor removal, infinite volume, phase transition, BCC selection, T6, and T7 are not claimed.